

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech****Branch : Chemical Engineering****Semester : VI****Subject Code & Name: BTHM604 Process Economics & Project Management****Max Marks: 60****Date: 26/12/24****Duration: 3 Hr****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Working capital is also known as____ (A) Operation capital (B) Operating capital (C) Current assets capital (D) Capital relating to main projects of the comp	CO1	1
2	Working capital = (A) Core current assets less current liabilities (B) Core current assets less core current liabilities (C) Liquid assets less current liabilities (D) Current assets less current liabilities	CO1	1
3	Time value of money indicates that A) A unit of money obtained today is worth more than a unit of money obtained in future B) A unit of money obtained today is worth less than a unit of money obtained in future C) There is no difference in the value of money obtained today and tomorrow D) None of the above	CO2	1
4	The main objective of depreciation is (A) To show the previous profit (B) To calculate net profit (C) To reduce tax (D) To satisfy the tax department	CO2	1
5	Depreciation is generated due to (A) Increase in the value of liability (B) Decrease in capital (C) Wear and tear (D) Decrease in the value of assets	CO3	1

6	According to straight line method of providing depreciation, the depreciation (A) Remains constant (B) Increase each year. (C) Decrease each year (D) None of them.	CO3	1
7	Salvage value means (A) Definite sale price of the asset (B) Cash to be received when life of the asset ends (C) Cash to be paid when asset is disposed off (D) Estimated disposal value	CO4	1
8	What is the basis for PERT analysis? (A) An optimistic time (B) A pessimistic period of time (C) The date that is most likely (D) All options mentioned above	CO4	1
9	What is the rule for the earliest start time? (A) It compares the activity's start time to that of a successor activity. (B) It compares the activity's end time to that of a previous activity. (C) It specifies when a project may begin. (D) It establishes the start date for a project.	CO3	1
10	What is the definition of a critical path? (A) It's a path that connects the starting and ending nodes. (B) It combines all of the paths. (C) It is the shortest route. (D) It is the most direct route.	CO4	1
11	Which of the following layout manufactures part in small or medium batches? A) Fixed Position Layout B) Mixed Position Layout C) Process Layout D) Product Layout	CO3	1
12	Which of the following factors affects plant location? (A) Availability of labour (B) Availability of fuel (c) Availability of power (D) All of these	CO4	1

Q. 2	Solve the following.	CO1	12														
A)	Explain Working capital	CO1	6														
B)	Explain fixed capital investment items for a chemical industry		6														
Q.3	Solve the following.		12														
A)	What is Interests and Annuity?	CO2	6														
B)	What is Time value of money?	CO2	6														
Q. 4	Solve Any Two of the following.		12														
A)	What is depreciation?	CO3	6														
B)	Rs.10000 is borrowed of an interest rate of 30%. Determine 1) Total amount of principal with simple interest after 5yr. 2) Total amount of principal with compound interest after 5yr.	CO3	6														
C)	A heat exchanger (HE) has been design. A HE with negligible scrap value cost Rs. 4000 will have useful life of 6yr. Another HE cost of Rs. 6800 with 10yr life and a scrap value Rs. 800. Assuming an effective compound interest rate of 8% per yr, determine which HE is cheaper by comparing capitalized costs .	CO3	6														
Q.5	Solve Any Two of the following.		12														
A)	A pump installation costing of Rs. 70000 with 7 yr life with no salvage value required Rs. 5000 yearly for maintenance and operation. What is the present worth of the service rendered by pump if the money worth is 13%.	CO3	6														
B)	Explain breakeven point in engineering economics	CO3	6														
C)	What is project evaluation and assessment for the project profitability?	CO3	6														
Q. 6	Solve Any Two of the following.		12														
A)	Construct a network diagram and find critical path: <table border="1"><tr><td>Activity</td><td>1-2</td><td>1-3</td><td>2-4</td><td>3-4</td><td>2-5</td><td>4-5</td></tr><tr><td>Duration</td><td>13</td><td>12</td><td>2</td><td>8</td><td>15</td><td>2</td></tr></table>	Activity	1-2	1-3	2-4	3-4	2-5	4-5	Duration	13	12	2	8	15	2	CO4	6
Activity	1-2	1-3	2-4	3-4	2-5	4-5											
Duration	13	12	2	8	15	2											
B)	What are the uses of PERT and CPM in project management	CO4	6														
C)	What are the important factors for the selecting a factory or plant site?	CO4	6														
	*** End ***																